Immersive Multi-Domain Decision-Making Trainer

SIH Problem Statement 26248 | Combined Final Concept

Purpose

A web-based, multi-participant simulation and decision-making training environment for degraded communication conditions. This document combines the 12 core solution points with additional concepts interpreted from the handwritten notes, including military map symbology, entity hierarchy, configurable behaviour, information visibility, and exercise review.

Part I — The 12 Core Solution Points

1. **Own Side and Enemy Side:** Represent both sides within the same simulation.
2. **Instructor / Control Staff:** Allow an instructor to manage the training exercise and its scenarios.
3. **Central Server:** Coordinate the simulation and maintain a shared state for participating users.
4. **Communication Disruption:** Simulate degraded communication between participants.
5. **Scenario Generation:** Create configurable training scenarios with different conditions.
6. **System-Generated Entities (SGEs):** Represent entities whose behaviour follows predefined or configurable behavioural patterns.
7. **GIS and 3D Maps:** Represent the simulated environment using maps, geographic layers, terrain, and 3D visualisation.
8. **Military Symbolisation:** Represent entities using military symbols on the map.
9. **Information Management:** Show participants the information available to them under the simulated communication conditions.
10. **Decision-Making Support:** Enable participants to make decisions using the information available to them.
11. **Algorithm and Error Analysis:** Examine system behaviour, possible errors, and the effects of incomplete information.
12. **Training Evaluation:** Record decisions and review the exercise to understand how communication disruption affects decision-making.

Part II — Additional Concepts from the Handwritten Notes

1. Military Map Symbology

Use a searchable symbol gallery to find, preview, place, and edit standard map symbols. Support point symbols for entities and multi-point tactical graphics for lines, boundaries, and areas. Where supported, attributes may include affiliation, echelon/unit size, status, identifiers, and coordinates.

2. Entity Hierarchy and System-Generated Sub-Entities

Model a parent-child structure such as scenario/formation → platoon or equivalent unit → subgroup → individual simulated entities. Let a parent generate configurable child entities from templates, with type, count, starting position, and attributes. Keep counts configurable rather than treating handwritten examples as fixed requirements.

3. Resolution and Levels of Detail

Support an aggregate formation view, a subordinate-unit view, and—if needed—individual entities. Allow users to drill down and return to the overview, while reducing map clutter when zoomed out.

4. Behaviour Patterns, Movement Tracks, and Events

Assign configurable virtual behaviour profiles to entities. Record timestamped positions and render movement tracks on the map. An event engine can evaluate configured triggers, notify the appropriate role, and record expected versus actual outcomes for review.

5. Role-Based Views and Information Flow

Keep the simulation's internal state separate from each participant's view. The instructor may access the full scenario state, while participant views depend on role and information availability. An entity may exist even when its report is delayed, missing, or unavailable to a participant. Log when information is created, delivered, and to whom.

6. GIS, Map Layers, and 3D Terrain

Start with a 2D GIS/base map, active layers, and annotations. Store points, lines, and areas as geographic features. Add 3D terrain or scene visualisation if needed, using the same scenario data.

7. Map Annotations and Scenario Workspace

Support text, pins, lines, rectangles, and areas in addition to standard military symbols. Keep symbol definition, entity identity, map placement, and annotation geometry as separate concepts.

8. Central Simulation Service

Coordinate the shared simulation clock and authoritative entity state. Maintain role-based participant sessions and permissions. If "FOV" means field of view, it may be treated as a visibility rule, but this needs stakeholder confirmation.

9. Decision Records and After-Action Review

Allow a trainee to record a decision and brief rationale at a point in the timeline. Capture what information was available at that time. Let the instructor review the timeline, entity tracks, events, and decisions using a configurable training rubric.

10. Error and Information Analysis

Record event triggers, information delivery, participant actions, and expected versus actual outcomes. Use these records to examine discrepancies and understand how incomplete or delayed information influenced decisions.

Part III — Suggested End-to-End Workflow

1. Prepare a GIS/base map, layers, and annotations.
2. Search the symbol gallery and place a point symbol or tactical graphic.
3. Configure supported symbol attributes and map position.
4. Build the parent-child entity hierarchy and generate child entities from templates.
5. Assign behaviour profiles and capture timestamped tracks.
6. Run the exercise through the central simulation service with role-specific information views and configurable communication disruption.
7. Record decisions, events, and information delivery, then review the exercise timeline and outcomes.

Suggested Data Concepts

Scenario entity: ID, type, display name, parent ID, attributes, status.

Symbol definition: standard, symbol code/name, supported modifiers.

Map placement: entity ID, coordinates/geometry, layer ID, label visibility.

Track point: entity ID, timestamp, coordinates, simulated status.

Behaviour profile: profile ID, type, parameters, schedule.

Information event: source, creation time, recipient role, delivery time, state.

Items to Confirm

Some handwritten terms and figures are uncertain, including “SAE”, “12 platoon size”, “50 tracks”, a possible “12 km”, “FOV”, and “20 TP”. Confirm their meaning with the stakeholder before treating them as fixed requirements. Also confirm the required symbol standard/version and whether 3D/AR/VR is mandatory or optional.

Interpretation note

This document combines the two supplied PDFs. Interpretations of unclear handwriting are tentative and are not presented as confirmed stakeholder requirements. The 12 core points are retained from the summary document.